

Permutation Order

Time limit: 1.00 s

Memory limit: 512 MB

Let $p(n, k)$ denote the k th permutation (in lexicographical order) of $1 \dots n$. For example, $p(4, 1) = [1, 2, 3, 4]$ and $p(4, 2) = [1, 2, 4, 3]$.

Your task is to process two types of tests:

1. Given n and k , find $p(n, k)$
2. Given n and $p(n, k)$, find k

Input

The first line has an integer t : the number of tests.

Each test is either " $1 \ n \ k$ " or " $2 \ n \ p(n, k)$ ".

Output

For each test, print the answer according to the example.

Constraints

- $1 \leq t \leq 1000$
- $1 \leq n \leq 20$
- $1 \leq k \leq n!$

Example

Input	Output
6	1 2 3 4
1 4 1	1 2 4 3
1 4 2	1
2 4 1 2 3 4	2
2 4 1 2 4 3	2 4 5 3 1
1 5 42	42
2 5 2 4 5 3 1	